

William Park

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EDUCATION

University of Waterloo

Bachelor of Applied Science in Honours Electrical Engineering
Cumulative GPA: 91.09%

Waterloo, ON

Sept 2024 – Present

TECHNICAL SKILLS

Languages: SystemVerilog, Verilog, VHDL, Tcl, Bash, Assembly, Python, C/C++

Developer Tools: Git, Vivado, Quartus Prime, IcarusVerilog, Verilator, gem5, Yosys, OpenLane, GDB, Docker, Linux

Libraries: cocotb, pyserial

Protocols: SPI, I2C, CAN, UART, AXI4

WORK EXPERIENCE

University of Waterloo

Undergraduate Research Co-op - Prof. Hiren Patel

May 2026 – Present

Waterloo, ON

- Designed a hardware prefetcher for indirect memory access patterns in **RISC-V Vector (RVV)** workloads, **achieving speedups of up to 2.20× and 1.62×** over no prefetching and scalar indirect memory prefetching respectively on sparse linear algebra and graph kernels
- Built the full simulation infrastructure in **gem5** using **C++**, extending an RVV platform with custom cache hierarchies and prefetcher models
- Characterized indirect memory access patterns across RVV kernels to shape the prefetcher's pattern detection algorithm
- Implementing the prefetcher in **SystemVerilog** to quantify the design's area overhead against a baseline RVV system

UWASIC Design Team

TinyTracer Team Lead

Oct 2025 – Present

Waterloo, ON

- Leading a student team to design and tape out the first programmable ray-tracing ASIC on Tiny Tapeout
- Designed the instruction set architecture, defining scalar and vector instruction encodings, the scene data format, and how vector operations decompose into scalar operations
- Defined the system architecture from scene loading to frame rendering, producing a block diagram, memory map, and RTL module templates for each functional block
- Established inter-block communication interfaces, including ready/valid handshaking, endianness, and instruction and data widths, enabling independent RTL development

Kiwi Charge

Electrical Engineering Intern

Sept 2025 – Dec 2025

Toronto, ON

- Developed a ROS2-based teleoperation pipeline for a four-wheel differential-drive prototype, integrating joystick input and real-time motor command streaming with the **CAN** bus protocol
- Designed an electrical schematic for the prototype, including power distribution, motor driver interfaces, CAN bus communication, and sensor integration, enabling reliable bring-up and debugging

PROJECTS

R500

June 2025 - Present

- Designed and implemented a **RISC-V** processor in **SystemVerilog** on the Xilinx Artix-7 FPGA
- Reached **100% compliance** with the RV32I ISA based on the official RISC-V compliance test suite
- Achieved a throughput of **46.3 million instructions per second** using a **57 MHz** clock, along with a **CPI of 1.23** and a **branch prediction accuracy of 96.5%**
- Optimized performance through pipelining, forwarding, stalling, flushing, and gshare global branch prediction
- Created unit testbenches in SystemVerilog, employing directed tests and constrained random verification
- Automated simulation, synthesis, and static timing analysis in Vivado using **Tcl** scripts
- Designing L1 instruction and data caches with shared **AXI4** bus